



IBM watsonx Code Assistant

For Red Hat Ansible Lightspeed

Lab Guide

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Purpose

This document will guide you through how to conduct a successful lab for WCA Ansible. It's intended for anyone who is conducting a hands-on lab for WCA Ansible.

Getting Started: Lab Setup

Start an Ansible free trial

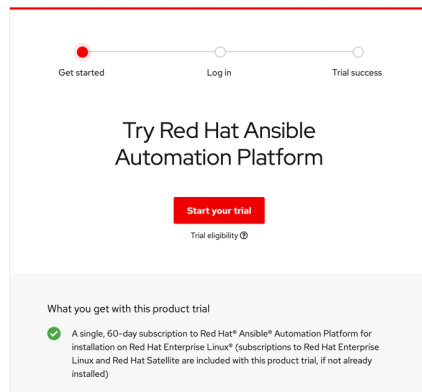
1. Visit the [get started webpage](#).
2. Login with your Red Hat credentials which will have been emailed to you.
3. Once you have logged in continue with the following steps:

Try Ansible LightSpeed with watsonx Code Assistant

To get started:

- If you're not currently an Ansible Automation Platform subscriber, sign up now for [no-cost 60-day trial](#) which includes access to the Ansible Lightspeed service.
- To start a trial of Ansible Lightspeed in Visual Studio Code, please refer to the [documentation](#).
- Refer to the Ansible Lightspeed [documentation](#) for more details.

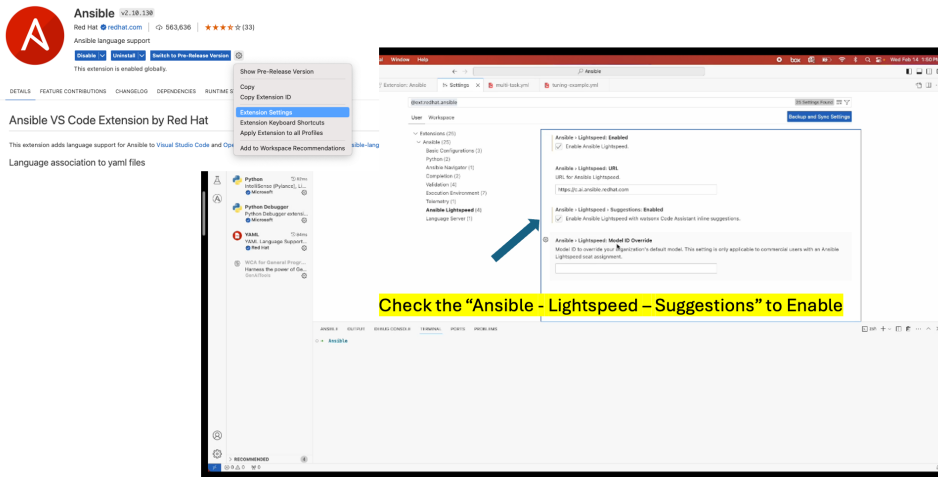
Key features of Ansible Lightspeed



Setup VS Code

1. Open Visual Studio Code.
2. Click on the Extensions view icon on the sidebar (or press Cntrl+Shift+X).
3. Search "Ansible".
4. Find the extension named "Ansible" (typically by Red Hat) and click Install.

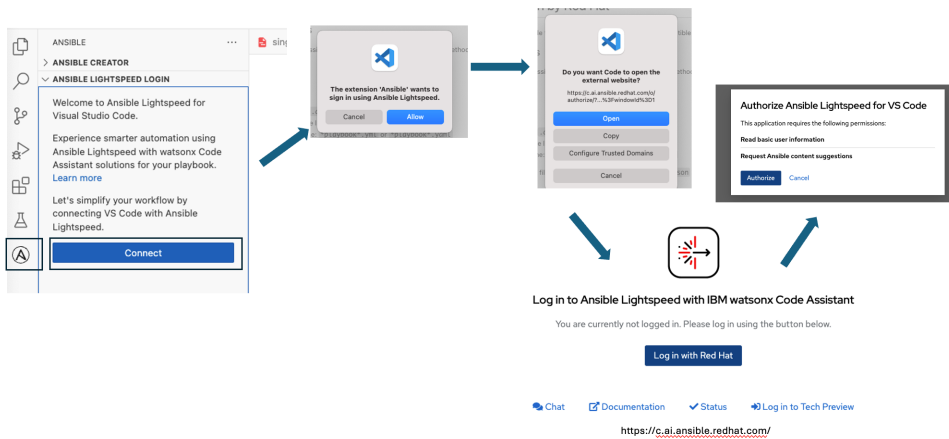
- Once the extension has been installed, open the extension settings and check 'Enable Ansible Lightspeed with watsonx Code Assistant inline suggestions' (see image below).



For more detailed instructions, watch the [interactive walkthrough](#).

Connect to Ansible Lightspeed login within VS Code

Please follow the steps below:



Ensure your License type is “Licensed”

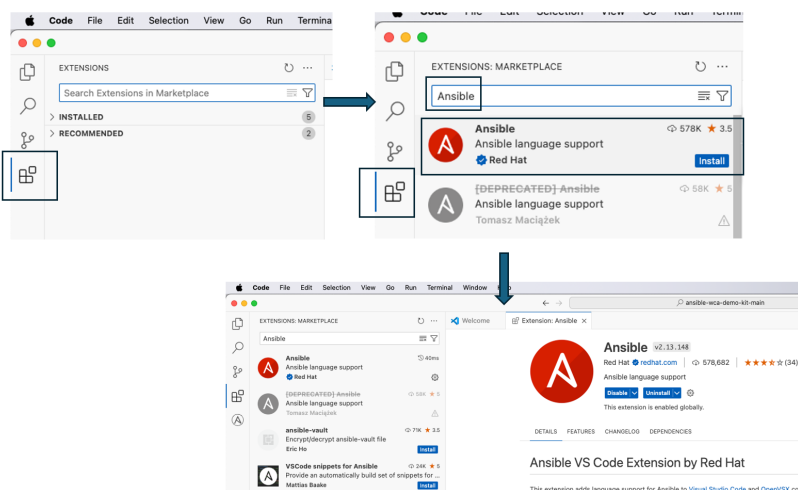
If you see “Unlicensed”, it could mean:

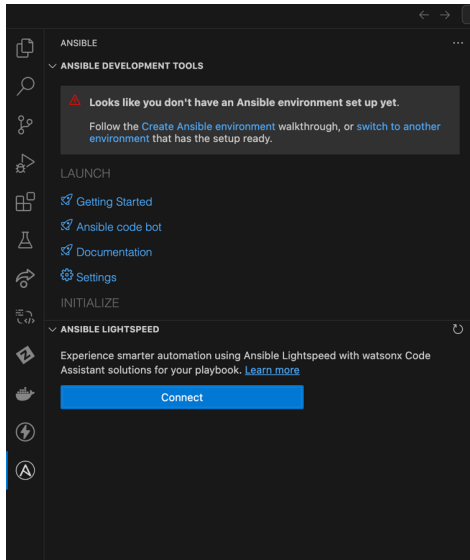
1. You are not using the Red Hat credentials provided to you for the lab. Or you have not verified your credentials yet.
Try this: Follow the instructions from the automated email from Red Hat you’ve received before the lab.
2. You are using your *other* Red Hat credentials. You can have multiple Red Hat IDs at once. Please use the credentials provided to you for the lab.
Try this: Logout from VS Code with your old Red Hat ID.

If problem persists, restart VS Code and wifi to reconnect to the internet.

Enable “Lightspeed” on Ansible Extension settings (VS Code)

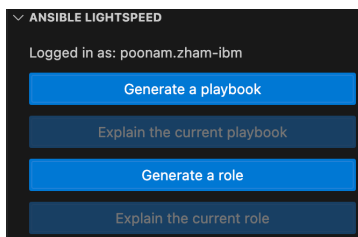
Please follow the steps below:



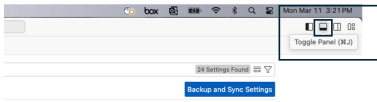


This should take you [here](#) to authorize.

Once connected it should look like the following image:



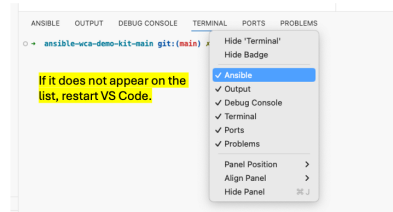
Confirm “**Ansible**” tab is visible on the Terminal panel



If the bottom terminal panel is missing, toggle the panel view on the top-right options on VS Code



Right click on Terminal

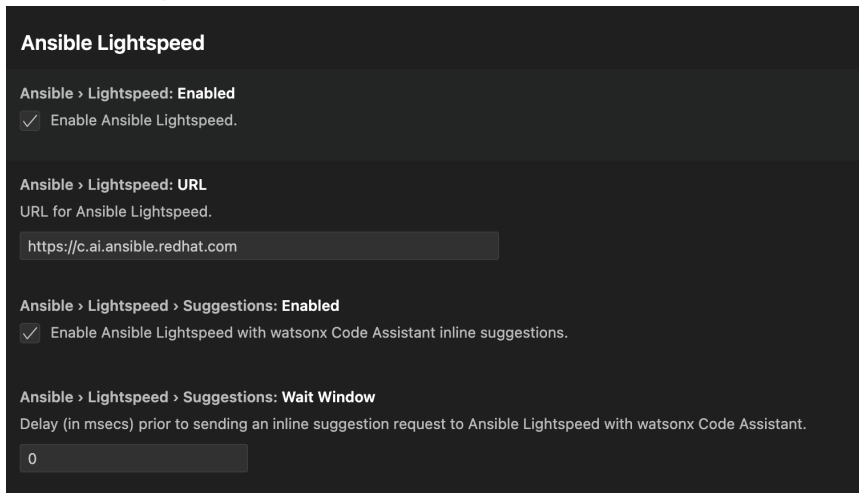


If it does not appear on the list, restart VS Code.

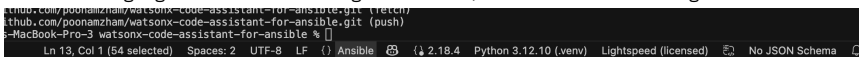
Fork [this repo](#).

Via Ansible extension settings

1. Verify if Ansible Lightspeed is enabled :

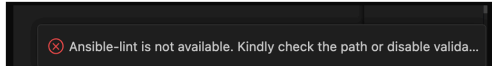


2. Make sure language in VSCode is showing as “Ansible”, and if not click and change it to “Ansible”:

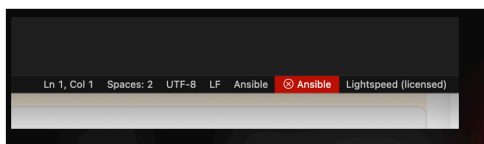


Ignore these warnings

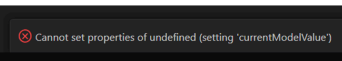
Once you open a file (Repo > Configure web server > single-task.yml), you would see some validation errors. They are not blockers to get WCA content suggestions, so please ignore them.



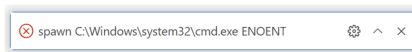
This warning is displayed due to Ansible validation. It constantly checks your Ansible syntax as you type. It is safe to IGNORE this error for the lab



The Ansible extension checks if you have Ansible installed locally. You can IGNORE this error as well.



Make sure the "Model ID Override" field is empty in your Ansible extension setting.



This warning is not related to Ansible. You can ignore this warning.

Setting Language Mode as “Ansible”

Prerequisite:

- Microsoft VS Code
- Ansible extension for VSCode
- Ansible
- ansible-lint
- python3
- GitHub account

Lab 1: Install and configure Apache end to end

1. Go to this [GitHub repo](#).
2. Navigate to 'Lab 1'.
3. Open the files and generate the code.
4. Go to the next line and wait once the code appears.
5. Click tab to accept the code.

6. Files to generate:

`incubation/Lab1_EndtoEndcompleteApacheInstall/Config/roles/httpd/tasks/main.yml`

`incubation/incubation/Lab1_EndtoEndcompleteApacheInstall/Config/roles/httpd/tasks/main.yml`

`incubation/Lab1_EndtoEndcompleteApacheInstall/Config/roles/httpd.conf/tasks/main.yml`

`incubation/Lab1_EndtoEndcompleteApacheInstall/Config/roles/index.html/tasks/main.yml`

Go to `inventory.ini` and add ip and user for server where you would like to install.

Command to run

`ansible-playbook -i inventory.ini apacheservers.yml --private-key <demoprivatekey>`

or

`ansible-playbook -i inventory.ini apacheservers.yml --ask-pass --user <user>`

The output should look like this:

```

• poonamzham@Poonams-MacBook-Pro-3 Config % ansible-playbook -i inventory.ini apacheservers.yml --ask-pass --user cecuser
SSH password:
PLAY [Top Level Apache Server Play] *****

TASK [Gathering Facts] *****
[WARNING]: Platform linux on host 129.40.96.65 is using the discovered Python interpreter at /usr/bin/python3.12, but future installation of another Python
interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.18/reference_appendices/interpreter_discovery.html for more
information.
ok: [129.40.96.65]

TASK [httpd : Create /ibmdata/htdocs directory root:root] *****
ok: [129.40.96.65]

TASK [httpd : Install httpd] *****
ok: [129.40.96.65]

TASK [httpd : Configure seport to listen on tcp 8080 and 8443 using community.general.seport] *****
ok: [129.40.96.65]

TASK [httpd : Enable httpd to start at boot] *****
ok: [129.40.96.65]

TASK [httpd : Open firewall ports 8080 & 8443] *****
changed: [129.40.96.65] => (item=8080/tcp)
changed: [129.40.96.65] => (item=8443/tcp)

TASK [httpd.conf : Create /var/log/httpd directory owned by root:root with 0750 permissions] *****
changed: [129.40.96.65]

TASK [httpd.conf : Create httpd.conf from a template and save as /etc/httpd/conf/httpd.conf] *****
changed: [129.40.96.65]

TASK [httpd.conf : Start httpd] *****
changed: [129.40.96.65]

TASK [index.html : Create index.html from a template and save it as /ibmdata/htdocs/index.html owned by apache:apache] *****
changed: [129.40.96.65]

RUNNING HANDLER [httpd.conf : Restart httpd] *****
changed: [129.40.96.65]

PLAY RECAP *****
129.40.96.65 : ok=11 changed=6 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0

```

To test run

`curl http://<ip>:8080`

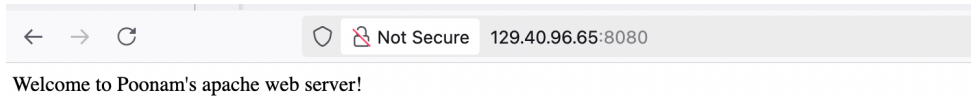
or go to `http://<ip>:8080`

Commented [KL1]: Needs formatting however I lack context to do correctly. Let's do via call?

Commented [PZ2R1]: Done



output:



Note: If its not running check if port is block due to firewall

Lab 2: Install and configure Cockpit

See: https://github.com/poonamzham/incubation/tree/main/Lab2_InstallandconfigureCockpitusingAnsible

Repeat steps same as in Lab 1

```
ansible-playbook -i inventory.ini install_cockpit_multi-task.yml --ask-pass --user <user>\n
```

Output:

Not Secure 129.40.96.65:9090

Red Hat Enterprise Linux

User name

This connection is not secure. Logins entered here could be compromised. [Learn More](#)

[Manage Passwords](#)

[Log in](#)

Server: p1333-pvm1

Log in with your server user account.

ceouser@p1333-pvm1

Limited access

Web console is running in limited access mode. [Turn on administrative access](#)

p1333-pvm1 running Red Hat Enterprise Linux 10.0 Beta (Coughlan)

Welcome to IBM Technology Zone

IBM's internal systems must only be used for conducting IBM's business or for purposes authorized by IBM management. Use is subject to audit at any time by IBM management.

Unauthorized access will be investigated and penalties will be pursued in conformance with applicable laws and regulations. If you are not an authorized user disconnect now.

IaaS Red Hat Enterprise Linux 10.0 Beta

Health

[Bug fix updates available](#)

[Not connected to Insights](#)

Last successful login: Apr 29, 11:31 PM
from :ffff:9.150.10.178 on web console
[View login history](#)

Usage

CPU 0% of 64 CPUs

Memory 2.3 / 9.3 GiB

System information

Machine ID: fb1c7586012644c9afa626cb54e9476d

Up since: 2 weeks ago

Configuration

Hostname: p1333-pvm1

System time: Apr 29, 2025, 9:33 AM

Domain: Install realmd support

Performance profile

Lab 3: Install and configure PostgreSQL and PGA Admin container

See:

https://github.com/poonamzham/incubation/tree/main/Lab3_InstallandconfigurePostgreSQLandPGAdmincontainer

Repeat steps same as in Lab 1 for

Or To generate playbook to install pgsq go to Lab 5

Prompt:

Install postgresql-server & Run postgresql setup command with postgresql.conf & Start and enable postgresql service

If you get privilege error add become:true

```
become: true
```

```
o-playbookgeneration / ❌ createpostgresql.yml
---
- name: Install postgresql-server and setup
  hosts: all
  become: true
  tasks:
    - name: Install postgresql-server
      ansible.builtin.package:
        name: postgresql-server
        state: present
    - name: Run postgresql setup command
      ansible.builtin.command: postgresql-setup --initdb
      args:
        creates: /var/lib/pgsql/data/postgresql.conf
    - name: Start and enable postgresql service
      ansible.builtin.service:
        name: postgresql
        state: started
        enabled: true
```



Lab 4: Customize your model and deploy on your WCA Ansible instance

See: <https://dataplatfom.cloud.ibm.com/wx/setup?context=wca>

IBM watsonx Code Assistant

Set up IBM watsonx Code Assistant for Red Hat Ansible Lightspeed Standard

Service instance: itzwca-665003hvcu-pu3qs0wo

Steps to complete

Refresh

Complete the following steps to enable AI-generated content recommendations in the Ansible VS Code Extension by Red Hat. Once you have completed a step, check the checkbox to move to the next step.

☒

Create a service ID and API key
Time to complete: 5 minutes
Create and download the API key. Use it to connect the Ansible VS Code Extension by Red Hat to watsonx Code Assistant.
[Learn more](#)

→

☒

Create deployment space and a model ID
Time to complete: 5 minutes
A deployment space is a serving environment for the base model. A model ID will be created after the deployment space has been created. The model ID determines which model will be used to generate content recommendations in the Ansible VS Code Extension by Red Hat.
[Learn more](#)
Model ID of base model
`b48e3c52-6135-408b-9a1e-8bcc75a47d32<|sepofid|>granite-3b`

→

☒

Add the service ID to the deployment space
Time to complete: 5 minutes
This will allow the Ansible VS Code Extension by Red Hat to interact with the deployment space using the API key.
[Learn more](#)

→

Follow only in case you want to perform model customization.